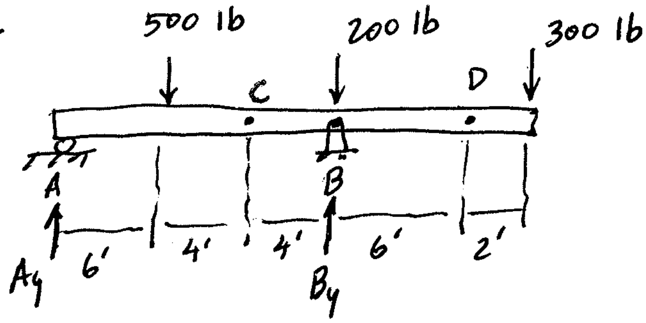
7-1



Find V and M at C and D.

Support Rxs: $+\circlearrowleft \Sigma M_B = 0;\quad A_y(14) - 500(8) + 300(8) = 0$

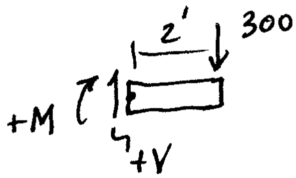
$$A_y = \frac{1600}{14} = 114.3 \text{ lb} \uparrow$$

$+\uparrow \Sigma F_y = 0;\quad A_y + B_y - 1000 = 0$

$$B_y = 1000 - 114.3$$

$$\boxed{B_y = 885.7 \text{ lb} \uparrow}$$

D

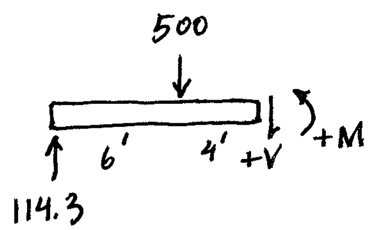


$\Sigma F_y = 0;\quad \boxed{V = 300 \text{ lb}}$

$+\circlearrowleft \Sigma M_D = 0;\quad +M + 300(2) = 0$

$$\boxed{M = -600 \text{ lb}\cdot\text{ft}}$$

C



$+\downarrow \Sigma F_y = 0;\quad 500 + V - 114.3 = 0$

$$\boxed{V = -385.7 \text{ lb}}$$

$+\circlearrowright \Sigma M_c = 0;\quad M + 500(4) - 114.3(10) = 0$

$$\boxed{M_c = -857 \text{ lb-ft}}$$